

National Research University Higher School of Economics

Faculty of Computer Science

Bachelor Programme “Data Science and Business Analytics”

Course Project

Development of an Alice Voice Assistant Skill for International Students of HSE

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Introduction

Voice assistants, like Alice (Yandex), Marusya (VK), Salute (Sber) and others have become an integral part of the modern artificial intelligence ecosystem, doing it accessible by using natural voice interaction patterns. The strong support for the Russian language, high-quality speech recognition, and deep integration with Yandex service infrastructure allows Alice to take a strong leading position among voice assistants in the Russian Federation. The Yandex. The dialogs platform allows developers to create what are sometimes called "skills," which are server-side applications connected via webhook that manage user requests and replies in the voice assistant environment.

International applications are a critical category in the university admissions system. They face even more challenges (opposite to local applicants) such as, language barriers, time zone differences and again specific Administrative and Educational Pathways within the Russian Federation. General admission-related information is usually found in various official documents and web pages, making it hard for international applicants to read between the lines and arrange necessary details on their own.

The Higher School of Economics annually attracts a significant number of international applicants, which increases the workload of the admissions office, especially during peak admission periods. Traditional communication channels—such as the official website, email correspondence, and phone consultations—do not always provide timely and convenient responses for users located outside the Russian Federation.

Under these conditions, the development of a specialized voice assistant skill for Alice, targeted specifically at international applicants of HSE, represents a relevant and practical solution. The use of a Retrieval-Augmented Generation (RAG) architecture, which retrieves relevant fragments of official documents before generating a response, allows the system to preserve factual consistency and minimize the risk of hallucinated or outdated information by grounding answers in official university sources.

Project Goal

The goal of this project is to develop a voice assistant skill titled "*HSE International Applicant*" for the Alice voice assistant, which provides international applicants of the Higher School of Economics with up-to-date and reliable information about the university admission process.

Project Objectives

To achieve the stated goal, the following objectives were defined:

1. Analyze the informational needs of international applicants and identify typical inquiry scenarios.
2. Design a system architecture based on the RAG approach, including a retrieval module, knowledge index, and response generation module.
3. Construct and structure a knowledge base derived from official HSE admission documents and web pages.
4. Perform data preprocessing, including text cleaning, chunk segmentation, and meta-data annotation.
5. Implement vector-based document indexing using embedding models.
6. Investigate the influence of retrieval parameters (e.g., top-k) on contextual completeness and relevance.
7. Compare the effectiveness of vector-based retrieval and keyword-based search methods.
8. Develop fallback strategies for cases of low-confidence retrieval or missing relevant documents.
9. Evaluate system responses in terms of factual grounding, completeness, robustness against hallucinations, and response latency.
10. Conduct system testing and formulate recommendations for further improvement and scalability.

Literature Review

Voice assistants represent a rapidly developing field within artificial intelligence, combining speech recognition, natural language processing, and speech synthesis technologies. In the educational domain, such systems are increasingly used to automate responses to frequently asked questions and to optimize communication between users and institutions.

With the advancement of large language models (LLMs), the issue of hallucination—generation of factually incorrect or fabricated information—has become a critical challenge. In domains that require strict adherence to official regulations, such as university admissions, such errors are unacceptable. To address this problem, the Retrieval-Augmented Generation (RAG) architecture was proposed. RAG combines a document retrieval component with a generative language model, where the model produces responses conditioned on retrieved evidence from an external knowledge base.

The RAG approach constrains generation to verified document fragments, thereby significantly improving factual grounding and reducing the likelihood of hallucinated content. Recent extensions of RAG include hybrid retrieval (combining vector similarity and keyword filtering), query expansion techniques, and confidence-based fallback strategies. These mechanisms are particularly relevant in high-stakes informational domains, including legal, medical, and academic admission systems.

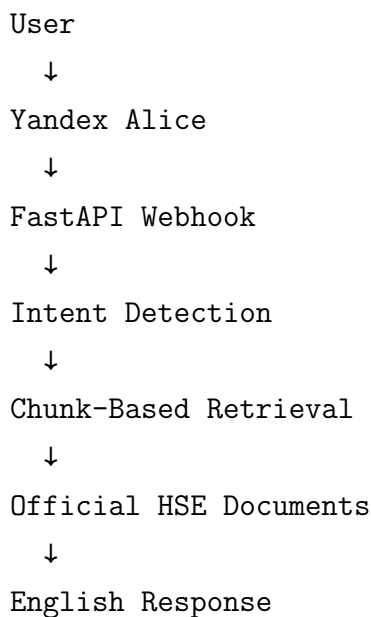
Within the Russian technological ecosystem, the Yandex.Dialogs platform provides a ready-to-use infrastructure for developing voice assistant skills. Leveraging an existing platform substantially reduces technical complexity compared to building a standalone voice assistant, which would require implementing a complete technology stack (Speech-to-Text, NLP, and Text-to-Speech) as well as dedicated cloud infrastructure.

An analysis of existing informational tools for HSE applicants reveals the absence of a specialized voice-based solution grounded in official university documents. This confirms the relevance of developing a voice assistant skill that integrates Yandex.Dialogs infrastructure with a RAG-based architecture to provide accurate, grounded, and accessible informational support for international applicants of the Higher School of Economics.

1 System Architecture

The developed assistant follows a lightweight Retrieval-Augmented Generation (RAG)-style architecture designed for international students of HSE University. The system combines document retrieval, intent detection, and response generation within a cloud-based deployment pipeline integrated with Yandex Alice.

The overall architecture of the system is presented below:



The user interacts with the assistant through Yandex Alice. User queries are forwarded to a FastAPI webhook deployed in the cloud. The webhook processes incoming requests, detects the query intent, searches the internal document collection, and generates a response in English.

The knowledge base consists of official HSE documents in DOCX format related to:

- migration procedures;
- dormitory and accommodation rules;
- admission regulations;
- university policies;
- international student support.

Before retrieval, all documents are preprocessed and split into overlapping text chunks. Chunk-based retrieval was selected because searching entire documents often produced noisy and weakly relevant results.

The retrieval module combines keyword matching and intent-aware scoring. Each user query is classified into one of several categories, including migration, dormitories, support, and university regulations. Text chunks are then ranked according to query relevance and intent-specific keywords.

The response generation component produces short English-language answers together with references to the corresponding HSE document sources. The system also includes simple off-topic filtering to reject questions unrelated to HSE or international student support.

The architecture was intentionally simplified to ensure stable deployment on free cloud infrastructure with limited computational resources.

2 Experimental Evaluation

2.1 Dataset

The system was evaluated on a manually constructed gold dataset consisting of 50 questions related to admission procedures for international applicants to HSE University.

Each dataset entry contains:

- the user question,
- the identifier of the relevant source document,
- the reference answer (used for manual validation).

The document corpus includes 10 official admission-related documents in `.docx` format. After preprocessing and chunking, the corpus produced 314 text chunks used for indexing.

2.2 Retrieval Configuration

ChromaDB was used as the vector storage and retrieval engine. Dense embeddings were generated using the `sentence-transformers/all-MiniLM-L6-v2` model.

The selected chunking configuration:

- Chunk size: 900 characters
- Overlap: 150 characters

For each query, the retriever returns the top- k most relevant chunks, where $k \in \{1, 3, 5\}$.

2.3 Evaluation Metrics

We use standard retrieval metrics:

- **Hit@k**: equals 1 if the correct document appears among the top- k retrieved results.
- **MRR@k (Mean Reciprocal Rank)**: evaluates how high the correct document appears in the ranking.

Mean Reciprocal Rank is defined as:

$$MRR = \frac{1}{N} \sum_{i=1}^N \frac{1}{rank_i}$$

where $rank_i$ denotes the position of the correct document for query i .

2.4 Baseline Retrieval Performance

The baseline configuration (MiniLM embeddings, chunk size 900/150) was evaluated on 50 gold questions.

	Hit@1	Hit@3	Hit@5	MRR@1	MRR@3	MRR@5
Baseline (900/150)	0.20	0.30	0.30	0.20	0.2467	0.2467

Table 1: Baseline dense retrieval performance

Chunk Size / Overlap	Hit@1	Hit@3	Hit@5	MRR@1	MRR@3	MRR@5
600 / 100	0.14	0.22	0.32	0.14	0.1733	0.1963
900 / 150	0.20	0.30	0.30	0.20	0.2467	0.2467
1200 / 200	0.12	0.20	0.20	0.12	0.1533	0.1803
1500 / 250	0.12	0.14	0.24	0.12	0.1300	0.1500

Table 2: Effect of chunk size on retrieval quality

2.5 Impact of Chunk Size

2.6 Discussion

The results indicate that medium-sized chunks (900/150) provide the best balance between contextual completeness and ranking precision.

Smaller chunks slightly improve Hit@5 but reduce ranking quality (lower MRR), indicating context fragmentation. Larger chunks decrease retrieval accuracy due to semantic dilution.

2.7 Conclusion of Experimental Study

The implemented dense retrieval system demonstrates baseline performance for admission-related queries:

- Hit@1 = 0.20
- Hit@3 = 0.30
- MRR@5 = 0.2467

Future improvements may include testing stronger multilingual embedding models, hybrid retrieval approaches (BM25 + dense retrieval), and reranking techniques to enhance ranking precision.

Project Repository:

<https://github.com/MikulaKula/Alice-applicants>

3 Experiments and Evaluation

3.1 Evaluation Methodology

To evaluate the retrieval quality of the RAG pipeline, a gold evaluation dataset consisting of 50 question-answer pairs was created manually based on official HSE documents for international students. Each record in the dataset contains:

- a user question,
- the reference answer,
- the source document,
- the supporting fragment,
- thematic category,
- difficulty level.

The retrieval system was evaluated using the following metrics:

- **Hit@1**,
- **Hit@3**,
- **Hit@5**,
- **MRR@5** (Mean Reciprocal Rank).

Hit@k measures whether at least one relevant chunk appears among the top-k retrieved chunks. MRR evaluates how highly the first relevant chunk is ranked.

The baseline configuration used:

- ChromaDB vector database,
- dense retrieval over document chunks,
- fixed-size chunking,
- retrieval-only pipeline without reranking.

3.2 Preprocessing Experiments

Several preprocessing strategies were evaluated in order to determine how text normalization affects retrieval quality. The following preprocessing methods were tested:

- raw text without preprocessing,
- basic preprocessing,
- strict preprocessing.

Table 3: Preprocessing Experiment Results

Method	Hit@1	Hit@3	Hit@5	MRR@5
Raw	0.20	0.30	0.30	0.2467
Basic	0.22	0.30	0.30	0.2567
Strict	0.22	0.28	0.30	0.2550

The results demonstrate that moderate preprocessing slightly improves retrieval quality. However, overly aggressive preprocessing may remove useful contextual information from official university documents. The best overall performance was achieved using the basic preprocessing strategy.

3.3 Chunking Experiments

Chunking strategy significantly affects retrieval quality in RAG systems. Several chunk sizes and chunking methods were evaluated. The following configurations were tested:

- fixed chunking with overlap,
- paragraph-based chunking.

Table 4: Chunking Experiment Results

Method	Hit@1	Hit@3	Hit@5	MRR@5
Fixed 600/100	0.14	0.22	0.30	0.1957
Fixed 900/150	0.20	0.30	0.30	0.2450
Fixed 1200/200	0.12	0.20	0.32	0.1803
Fixed 1500/250	0.12	0.14	0.24	0.1500
Paragraph 900/0	0.10	0.34	0.48	0.2343

The experiments showed that small chunks lose semantic context, while excessively large chunks reduce retrieval precision. The fixed chunk size of 900 characters with 150 overlap provided the most balanced performance overall.

Paragraph-based chunking achieved the best Hit@5 score because larger logical sections more frequently contained the correct answer. However, this method reduced Hit@1 precision.

3.4 Reranking Experiments

To improve retrieval quality, an additional lexical reranking stage was tested. The reranker re-ordered retrieved chunks using lexical similarity scores after dense retrieval.

Table 5: Reranking Experiment Results

Method	Hit@1	Hit@3	Hit@5	MRR@5
Retrieval only	0.20	0.28	0.30	0.2450
Retrieval + lexical reranker	0.20	0.28	0.30	0.2417

The lexical reranker did not improve retrieval quality. The results remained nearly identical to the baseline retrieval-only approach. One possible explanation is that official university documents contain highly formalized vocabulary and repetitive terminology, which are already captured effectively by dense retrieval methods.

3.5 Generation Evaluation

The generation component was evaluated manually. For selected questions from the gold dataset, answers were generated based on retrieved chunks and then evaluated according to the following criteria:

- factual correctness,
- relevance,
- completeness,
- absence of hallucinations,
- consistency with official HSE information.

Manual evaluation was selected because factual accuracy is critically important for university-related information systems. Incorrect or hallucinated answers may negatively affect international students during their adaptation process.

The evaluation demonstrated that retrieval quality strongly influences generation quality. Most generation errors were caused by incomplete or partially irrelevant retrieved chunks rather than by the generation process itself.

3.6 Discussion of Results

The experiments demonstrate that retrieval quality in educational RAG systems strongly depends on preprocessing and chunking strategies.

Moderate preprocessing improves retrieval stability, while excessive preprocessing may reduce contextual richness. Chunk size selection also plays a critical role. Medium-sized chunks achieved the best balance between semantic completeness and retrieval precision.

Reranking provided limited benefit in the current setup because the dataset consists primarily of structured official documents with repetitive terminology.

Overall, the developed RAG pipeline successfully retrieves relevant information from official HSE documents and can support international students in solving organizational and academic adaptation tasks.

One important limitation of the current experiments is the relatively low retrieval quality in some configurations. This behaviour is partially explained by several factors.

First, the document collection is relatively small and contains heterogeneous administrative documents with different writing styles and structures.

Second, the experiments involve English-language queries while most source documents are written in Russian. This multilingual mismatch negatively affects retrieval quality, especially for keyword-based approaches.

Third, the lightweight retrieval pipeline used in the deployed version does not include semantic reranking or advanced embedding optimization techniques.

Finally, some retrieved chunks contain broad administrative information rather than highly specific answers, which reduces precision for detailed user questions.

Despite these limitations, the experiments demonstrate that chunk-based retrieval significantly improves answer relevance compared to full-document retrieval and provides acceptable quality for a prototype university assistant.

3.7 Agentic RAG Experiment

An additional experiment was conducted using a simplified Agentic RAG approach. Instead of performing only direct retrieval, the system first selected an action depending on the user query type.

The Agentic RAG approach was selected instead of GraphRAG because the project domain mainly consists of procedural university documents, regulations, and FAQ-style information rather than highly interconnected entities and relationships. For this type of knowledge base, the main challenge is not graph traversal, but correct query routing, ambiguity handling, and filtering of irrelevant requests.

In addition, Agentic RAG is computationally simpler and easier to integrate into a lightweight university assistant system. The approach allows the system to dynamically decide whether retrieval is necessary, whether clarification should be requested, or whether the query should be rejected as off-topic. This behavior is especially important for real student interactions, where users often submit incomplete or unrelated questions.

The implemented agent supported several possible actions:

- `retrieve_answer` — perform standard retrieval from the vector database;
- `ask_clarification` — request clarification for ambiguous queries;
- `refuse_offtopic` — reject queries unrelated to the university domain.

Several test queries were used to evaluate the routing logic.

User Query	Selected Action
What documents do I need for migration registration?	retrieve_answer
Dormitory	ask_clarification
What is the weather today?	refuse_offtopic
Where can I find the contact email?	refuse_offtopic

Table 6: Agentic RAG routing behavior

The experiment demonstrated that the agent-based routing mechanism can improve robustness of the system by filtering irrelevant requests and handling ambiguous queries before retrieval. Compared to a standard RAG pipeline, the Agentic RAG approach provides an additional decision-making layer that improves interaction control and reduces unnecessary retrieval operations.

4 Deployment and Alice Skill Integration

The developed assistant was integrated with Yandex Alice using the Yandex Dialogs platform and deployed as a cloud web service.

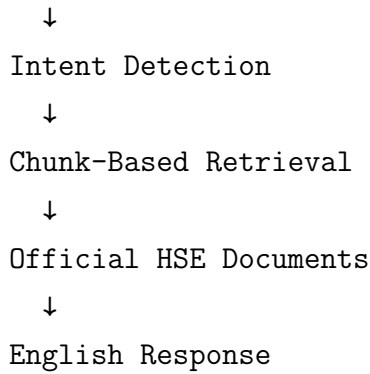
The backend was implemented using FastAPI in Python. The webhook receives HTTPS requests from Yandex Alice, processes user queries, searches relevant information in the HSE document collection, and returns English-language responses.

An example response generated by the webhook is shown below:

```
{
  "response": {
    "text":
      "According to the available HSE documents,
      questions about migration registration
      should be checked through the official
      university procedures for international students."
  }
}
```

The deployed architecture is shown below:

```
User
  ↓
Yandex Alice
  ↓
FastAPI Webhook
```



Initially, the system was tested locally using localtunnel and Cloudflare Tunnel. However, these solutions produced unstable webhook connections and timeout errors. As a result, the final production version was deployed on the Render cloud platform.

The deployed version uses lightweight chunk-based retrieval instead of full semantic vector retrieval. This design decision was made because the free Render infrastructure has strict memory limitations, which caused instability when using ChromaDB with dynamic embedding generation.

The retrieval pipeline works as follows:

1. Official HSE DOCX documents are loaded from the document collection.
2. Documents are cleaned and split into overlapping text chunks.
3. The system detects the query intent using predefined keyword rules.
4. Text chunks are scored according to keyword relevance and intent-specific terms.
5. The best matching chunk is selected.
6. The assistant returns a short English response together with the relevant source document.

The deployed assistant currently supports several categories of international student questions, including:

- migration registration;
- dormitories and accommodation;
- university regulations;
- support and contact information.

The system also includes simple off-topic filtering. If the question is unrelated to HSE or international student support, the assistant informs the user that no relevant information was found in the official documents.

Although the current deployed version does not use full semantic embeddings in production, experiments with ChromaDB and vector retrieval were conducted locally. These experiments demonstrated the potential for future improvements in retrieval quality and semantic search capabilities.

4.1 Production and Experimental Configurations

Two different system configurations were used during the project.

The experimental configuration was designed for retrieval evaluation and local testing. It used ChromaDB together with semantic embeddings and dense vector retrieval. This configuration allowed experiments with chunking strategies, preprocessing methods, reranking, and semantic search quality evaluation.

However, the experimental pipeline required significantly more memory and computational resources. During deployment testing on the free Render infrastructure, dynamic embedding generation and ChromaDB indexing caused memory overflows and unstable service behaviour.

As a result, the final deployed production version uses a lightweight retrieval pipeline based on:

- chunk-based document splitting;
- keyword matching;
- intent-aware scoring;
- template-based response generation.

Although the lightweight production configuration is less semantically powerful than the experimental ChromaDB pipeline, it provides stable cloud deployment and acceptable response quality for the prototype assistant.

This separation between experimental and production configurations demonstrates the engineering trade-off between retrieval quality and deployment constraints in low-resource cloud environments.

5 Future Work

The current prototype demonstrates a working integration of Yandex Alice with a lightweight RAG-style retrieval pipeline for international student support.

However, several improvements may be implemented in future versions of the system.

First, the document collection may be expanded with additional official HSE materials related to migration, dormitories, and university procedures.

Second, the retrieval pipeline may be upgraded with semantic embeddings and hybrid retrieval methods to improve answer relevance and multilingual search quality.

Another possible improvement is adding reranking models and better source selection mechanisms.

The assistant may also be extended with dialogue memory and support for more complex conversational scenarios.

Despite the current limitations, the developed prototype demonstrates the practical feasibility of deploying a university-oriented voice assistant based on Retrieval-Augmented Generation principles.